AttendWise

ALTERNATIVE ASSESSMENT TEST – 1 · 2026-27

Smart attendance, considered.

Portfolio-driven development report for a student-first attendance planner that helps every class decision feel more informed.

Student name _____

USN _____

Semester / section _____

Subject _____

Project title
AttendWise — Smart Attendance Planner

Repository
github.com/K-P-Videsh-Kumaar/Attendwise

Problem, purpose & objectives

AttendWise turns attendance from a static college-portal percentage into a clear, private planning system for students.

Problem statement

Students often discover that an attendance threshold is at risk when there is little room left to recover. Existing portals display historic percentages but do not explain the impact of future choices. AttendWise provides a student-managed planning view without claiming to replace official institutional records.

Six product objectives

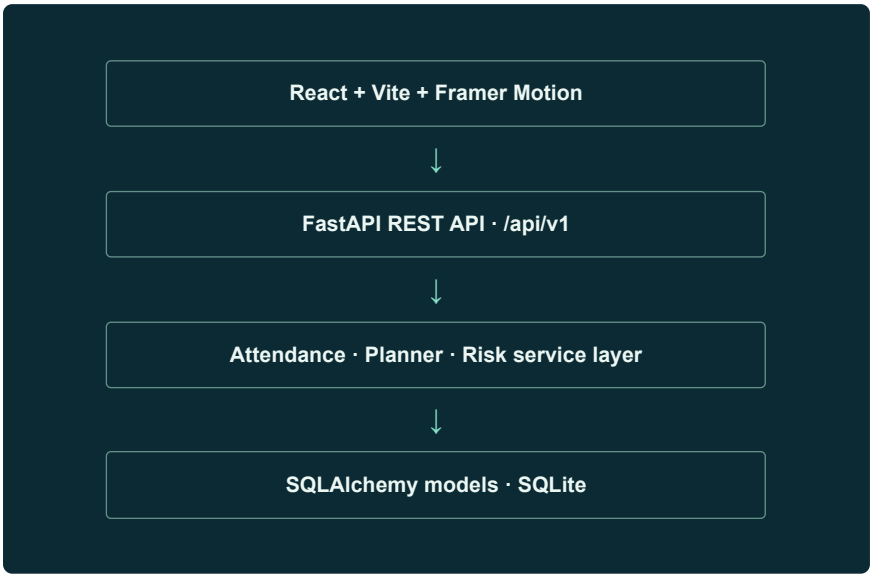
1. Secure student access
2. Subject-level targets
3. Accurate class logging
4. Clear risk indicators
5. Read-only scenario planning
6. Practical recovery guidance

Technology & methodology

Layer	Technology	Purpose
Interface	React + Vite	Responsive single-page experience.
Motion	Framer Motion	Purposeful page, card and progress feedback.
API	FastAPI + Pydantic	Typed REST routes and validation.
Data	SQLAlchemy + SQLite	Portable persistence.
Quality	Pytest + Playwright	Rule validation and browser evidence.

Architecture & key rules

Business rules are server-owned. The interface requests validated answers; it never treats browser-side calculation as authoritative.



Attendance

A / T × 100

A = attended classes.
T = attended + absent classes.
Cancelled records are excluded.

Recovery

ceil((r × T − A) / (1 − r))

The planner returns zero when on target and handles 100% as a special case.

Read-only guarantee. Scenario inputs return a projection only. No hypothetical value is ever written to the attendance register.

Important code excerpts

Centralised attendance calculation

```
attended = baseline_attended + attended records; total = attended + absent; percentage = round(attended / total * 100, 2) if total else None
```

> The calculation service excludes cancelled classes and produces no percentage when there are no counted classes.

Safe, server-side simulation

```
subject = owned_subject(subject_id, user, db); before = subject_metrics(subject, db); after = simulate(before, attended, missed); return {"read_only": True, "projections": result}
```

> The endpoint verifies ownership, calculates the result and makes no data write.

Information-led motion

```
<motion.i initial={{ width: 0 }} animate={{ width: `${percentage}%` }} transition={{ duration: .8 }} />
```

> Framer Motion draws each progress bar to a real value, making the percentage easier to perceive without adding visual noise.

Development approach

Model the domain. Users, subjects and class records use explicit status and duplicate-session rules.

Protect data. Every protected endpoint verifies ownership server-side.

Design for clarity. The interface uses hierarchy, risk colour and restrained motion to make status easy to scan.

Output screenshots

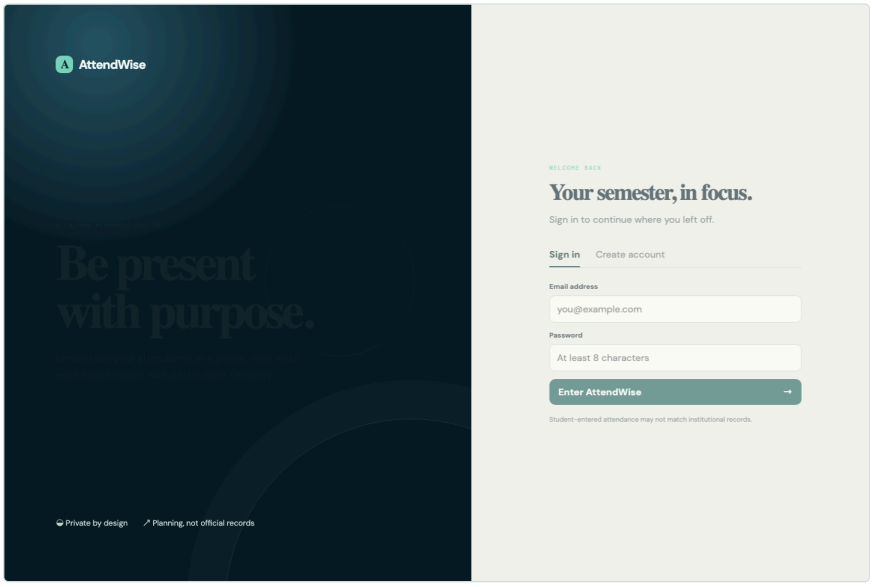


Figure 1 — Sign-in experience: a focused product proposition and clear account entry.

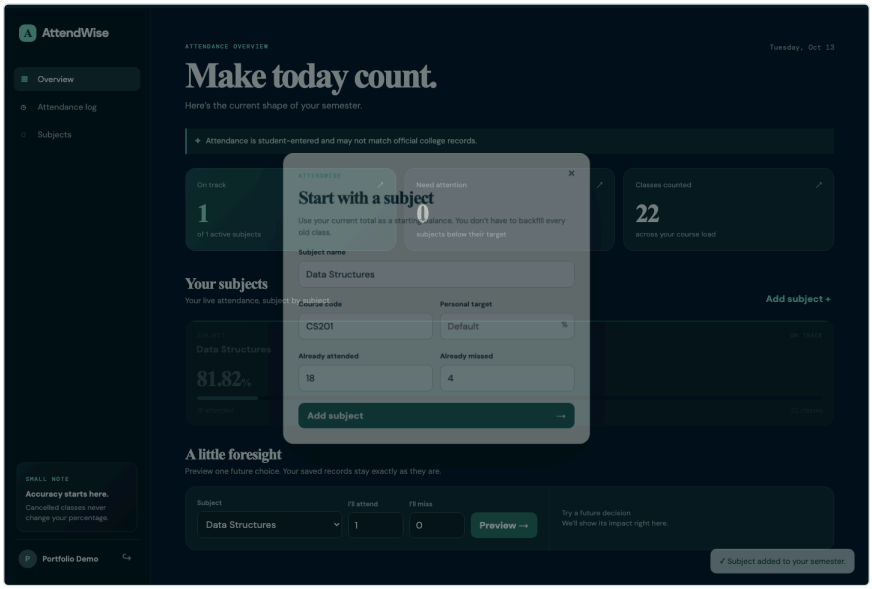


Figure 2 — Dashboard: live subject status and the read-only what-if planner.

Mobile output, learning & conclusion

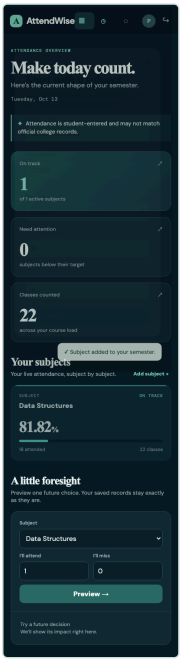


Figure 3 — Mobile dashboard preserving navigation, subject state and planning controls.

Challenges addressed
Cancelled records, no-data states, data ownership and non-mutating simulation required explicit backend rules.

Learning outcomes
FastAPI services, authentication, tested calculations, React integration, Framer Motion and browser-based validation.

Conclusion
AttendWise meets the MVP goal: students can create subjects, record outcomes, understand risk, preview future attendance and plan recovery. It demonstrates secure backend rules paired with a responsive, refined frontend.

Student declaration
I hereby declare that the project submitted as part of AAT-1 is my work and that I understand its implementation and contents.

Student name & signature

Date